

Supplementary Material: Regime-Conditioned Dynamical-EVT Diagnostics for Predictive Maintenance

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1 Extended Methods

The implemented state-space construction requires causal availability. If a delay embedding is used, every lagged coordinate is shifted so that Z_t contains only information observed at or before t . Scaling parameters, dangerous-region prototypes, and thresholds are fitted on the declared training or calibration partitions and reused without refitting on the test partition.

Runs declustering uses a non-negative integer run length m . If exceedance times are $e_1 < \dots < e_k$, then e_i and e_{i+1} belong to the same cluster when $e_{i+1} - e_i \leq m + 1$. Cluster severity may be summarized by size, duration, maximum observable, or integrated excess above threshold. These empirical quantities should not be equated with fault duration unless the sampling frequency, gap rule, and label definition justify that interpretation.

The inter-exceedance estimator is included because cluster inference can be sensitive to manual declustering choices [2]. The dynamical interpretation follows the hitting-time connection between returns to shrinking target sets and extreme-value laws [3]. The periodicity result motivates the caution that recurrence can reduce the extremal index without implying a specific mechanical cause [4].

Horizon-risk calibration is computed after freezing the event target. The current generated reliability diagram is a diagnostic based on ranked risk surrogates; a full probabilistic calibration claim would require held-out calibration data, uncertainty bands, and explicit acceptance criteria.

2 Additional Simulations

The simulation registry includes independent, periodic, noisy, mixture, missingness, and degradation families. Independent simulations check whether estimators approach the non-clustered target. Periodic simulations check whether the workflow detects clustered returns. Noisy simulations test whether finite perturbations mask recurrence, following the warning that observed noise can move finite samples toward apparent independence [1]. Regime mixtures and preprocessing perturbations test whether clustering can be generated by analysis choices rather than physical degradation.

The generated simulation asset used in the main paper is the reproducibility-scale artifact. It is sufficient for build validation, but it is not a replacement for the full registered experiment matrix. The supplement therefore treats simulation results as evidence about the implemented pipeline and estimator sensitivity, not as an exhaustive finite-sample theory.

2.1 Reference availability

Table S1 records, for every simulated family, how many configurations carry a reference extremal index, whether that reference is theoretical or numerical, and which clustering regime it falls in. Three families — noisy periodic dynamics, persistent shift, and regime mixture — have no established reference at any configuration: the two long-run estimates used to fix a numerical value disagree beyond the accepted tolerance, with discrepancies from 0.15 to 0.60 and no plateau in the block-based family. These are mixture and non-stationary processes for which a single extremal index is arguably not well defined, and they are reported as non-identifiable rather than assigned a value that one declustering convention happens to produce.

The complete estimator grid, carrying all thirteen reported quantities for every configuration and estimator, is distributed as `estimator_full_grid.csv` in the reproducibility package. Every value in the main-paper table is recomputable from it.

2.2 Sensitivity to perturbations

Figure S1 reports mean estimates under noise, missingness, and regime mixture. Added noise moves every estimator towards independence, which is the expected direction: perturbation breaks the deterministic recurrence that produces clustering, and the numerically established references for the noisy families sit between 0.94 and 0.98 accordingly. This is also why the high-replication grid, which runs at a single non-zero noise level, is reported in the main paper as evidence about near-independent processes.

Table S1: Reference availability and clustering regime for every simulated process family. Configurations without an established reference contribute no bias, RMSE or coverage evidence.

process	configurations	with reference	reference kind	reference theta	regime
burst	1	1	theoretical	1.000	near_independent
missingness					
cyclic	37	24	numerical	0.049, 0.062,	strongly_clustered
degradation				0.086, 0.175	
downsampling	1	1	numerical	0.981	near_independent
iid bounded	37	37	theoretical	1.000	near_independent
tail					
iid light tail	38	38	theoretical	1.000	near_independent
iid pareto	37	37	theoretical	1.000	near_independent
lagged	37	12	numerical	1.000	near_independent
multivariate					
logistic	37	31	numerical,	0.983, 0.984,	near_independent
nonperiodic			theoretical	0.992, 1.000	
target					
logistic	38	31	numerical,	0.500, 0.939,	moderately_clustered,
periodic			theoretical	0.940, 0.970	near_independent
target					
missing at	1	1	theoretical	1.000	near_independent
random					
noisy periodic	1	0	none		not
dynamics			established		established
persistent	1	0	none		not
shift			established		established
regime	37	0	none		not
mixture			established		established
smoothing	1	1	numerical	0.272	strongly_clustered

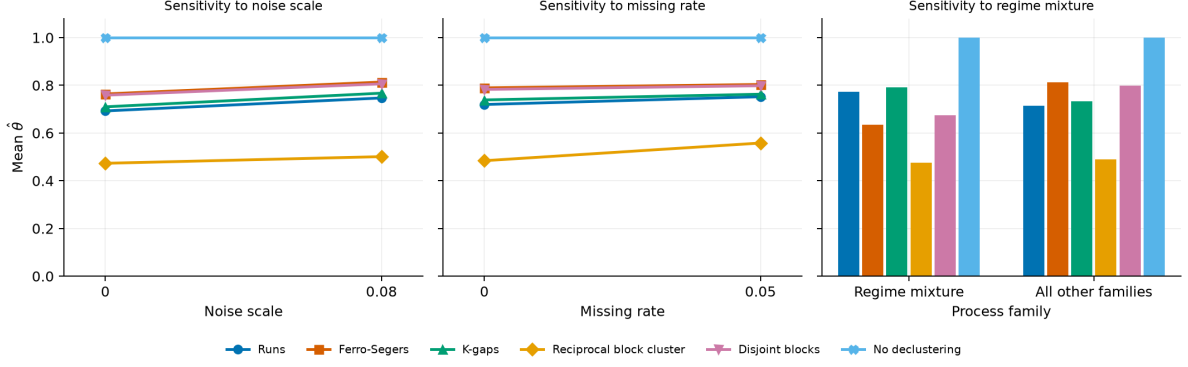


Figure S1: Estimator sensitivity to noise scale, missing rate, and regime mixture. The regime-mixture panel contrasts families rather than sweeping a parameter, since mixture is a property of the process and not a tuning knob.

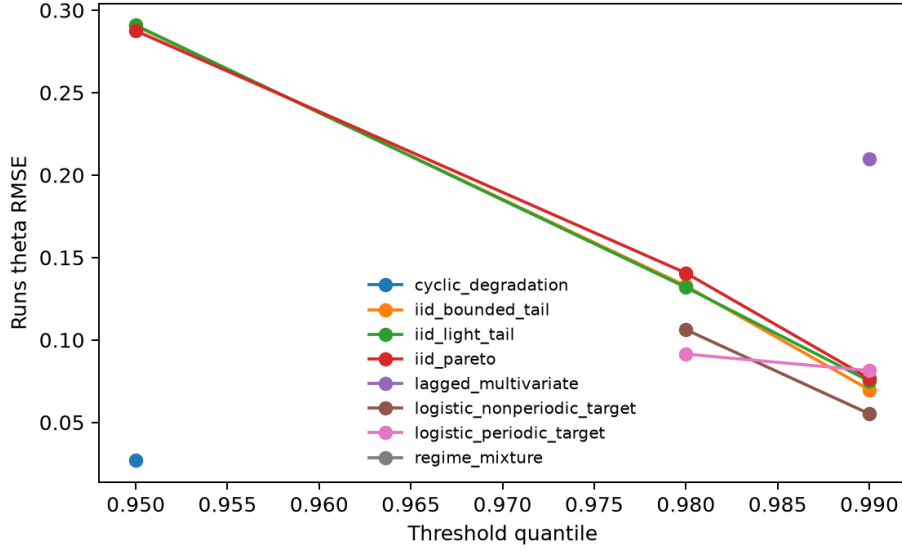


Figure S2: Supplementary copy of the generated simulation RMSE artifact.

3 Complete Event-Level Benchmark

Table S2 reports the complete held-out event-level benchmark under the shared alarm policy: every declared method on every compressor dataset, with detection status, event recall and precision, both alarm-burden axes, duplicate alarms and event utility. Table S3 records what each method actually computes, and Table S4 is the fairness audit.

Two properties of this benchmark are worth stating explicitly, because both were previously invisible.

First, method identity is enforced rather than assumed. Each method is declared by its score, its threshold rule and its event policy, and no two methods share all three. An earlier version of this benchmark reported nineteen methods of which nine produced bit-identical results, because a fallback assigned one shared score to any method lacking an explicit implementation; the classical peaks-over-threshold entry fitted no tail model and the declustering entries declustered nothing differently. The audit now reports any methods whose results coincide, so numerical agreement is visible as agreement rather than presented as independent evidence.

Second, alarm merging belongs to the detector. Four methods exist to vary declustering,

so requiring one event conversion across the benchmark would make them indistinguishable by construction. The shared elements are the warning horizon, the matching tolerance, the matching rule, the operating exposure and the independent unit; the merge gap each method used is reported per row.

Table S2: Complete held-out event-level benchmark under the shared alarm policy: every declared method on every compressor dataset. Alarm burden is given as both interruption frequency (false alarms per operating day) and warning exposure (percentage of the test period under warning), because the two can disagree in direction.

Data	Method	Family	Det.	Recall	Prec.	FA/day	Expo. %	Dup.	Utility
MetroPT	Single-sensor threshold	baseline	yes	1	0.000441	90.9	1.16	1	-452
MetroPT	Classical POT/GPD	baseline	yes	1	0.000395	101	1.27	3	-505
MetroPT	Nonlinear autoencoder	baseline	yes	1	0.000187	214	33.58	11	-1.07e+03
MetroPT	Conformal anomaly	baseline	yes	1	0.000394	102	2.70	3	-506
MetroPT	Empirical horizon risk	baseline	no	0	0	0	0.00	0	-2
MetroPT	Engineering threshold	baseline	yes	1	0.000387	103	14.88	3	-515
MetroPT	Ferro-Segers policy	baseline	yes	1	0.000392	102	0.12	3	-509
MetroPT	Fixed-run declustering	baseline	yes	1	0.000394	102	0.12	3	-506
MetroPT	Global empirical threshold	baseline	yes	1	0.000394	102	2.70	3	-506
MetroPT	Isolation forest	baseline	yes	1	0.000388	103	14.35	3	-514
MetroPT	K-gaps policy	baseline	yes	1	0.000392	102	0.12	3	-509
MetroPT	Linear autoencoder	baseline	yes	1	0.000212	189	7.35	5	-942
MetroPT	Regime-conditioned threshold	baseline	yes	1	0.000395	101	0.91	3	-504
MetroPT	Robust change point	baseline	yes	1	0.000386	104	14.79	3	-516
MetroPT	SPOT	baseline	no	0	0	1.28	0.04	0	-8.4

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Data	Method	Family	Det.	Recall	Prec.	FA/day	Expo. %	Dup.	Utility
MetroPT	Negative-control region	negative control	no	0	0	7.74	0.30	0	-40.6
MetroPT	Recurrence score	registered	yes	1	0.000361	111	0.13	3	-552
MetroPT	Failure-prototype region	target region	no	0	0	0	0.00	0	-2
MetroPT	Rare-state region	target region	yes	1	0.00112	35.6	0.88	3	-177
MetroPT2	Single-sensor threshold	baseline	yes	1	0.00282	21.2	7.38	3	-69.3
MetroPT2	Classical POT/GPD	baseline	yes	1	0.00112	53.7	2.15	3	-176
MetroPT2	Nonlinear autoencoder	baseline	yes	1	0.00128	47.3	2.33	3	-155
MetroPT2	Conformal anomaly	baseline	yes	1	0.00113	53.3	2.89	3	-175
MetroPT2	Empirical horizon risk	baseline	no	0	0	0	0.00	0	-2
MetroPT2	Engineering threshold	baseline	yes	1	0.00163	37.2	21.26	0	-122
MetroPT2	Ferro-Segers policy	baseline	yes	1	0.00113	53.3	0.06	3	-175
MetroPT2	Fixed-run declustering	baseline	yes	1	0.00113	53.3	0.06	3	-175
MetroPT2	Global empirical threshold	baseline	yes	1	0.00113	53.3	2.89	3	-175
MetroPT2	Isolation forest	baseline	yes	1	0.000574	105	6.71	7	-346
MetroPT2	K-gaps policy	baseline	yes	1	0.00113	53.6	0.06	3	-176
MetroPT2	Linear autoencoder	baseline	yes	1	0.000864	70.2	18.14	0	-230
MetroPT2	Regime-conditioned threshold	baseline	yes	1	0.000424	143	10.93	3	-471
MetroPT2	Robust change point	baseline	yes	1	0.000737	81.9	6.94	6	-270

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Data	Method	Family	Det.	Recall	Prec.	FA/day	Expo. %	Dup.	Utility
MetroPT2SPOT		baseline	no	0	0	1.09	0.05	0	-5.6
MetroPT2Negative-control region		negative control	no	0	0	41.8	2.62	0	-140
MetroPT2Recurrence score		registered	yes	1	0.00112	54	0.06	3	-177
MetroPT2Failure-prototype region		target region	yes	1	0.333	0.121	16.42	0	0.6
MetroPT2Rare-state region		target region	yes	1	0.00145	41.8	17.69	0	-137

Table S3: Method provenance for the benchmark: score, threshold rule, event policy, alarm merge gap, parameter count, calibration status and tuning partition. Methods sharing a score are separated by their threshold rule or event policy; no two share all three.

Method	Score	Threshold rule	Event policy	Merge gap	Params	Calibration	Tuned on
Single-sensor threshold	best single sensor z	train quantile	merge gap	60	2	uncalibrated	train
Classical POT/GPD	robust max abs z	gpd return level	merge gap	60	3	uncalibrated	train
Nonlinear autoencoder	nonlinear projection residual	train quantile	merge gap	60	8	uncalibrated	train
Conformal anomaly	conformal tail probability	conformal level	merge gap	60	1	conformal (marginal, split)	train (calibration split)
Empirical horizon risk	Empirical horizon risk	train quantile	merge gap	60	2	empirical frequency	train
Engineering threshold	robust max abs z	fixed physical 3z	merge gap	60	1	uncalibrated	none (fixed a priori)
Ferro-Segers policy	robust max abs z	train quantile	ferro segers run length	0	2	uncalibrated	train
Fixed-run declustering	robust max abs z	train quantile	Fixed-run declustering	0	2	uncalibrated	train
Global empirical threshold	robust max abs z	train quantile	merge gap	60	1	uncalibrated	train
Isolation forest	Isolation forest	train quantile	merge gap	60	100	uncalibrated	train
K-gaps policy	robust max abs z	train quantile	k gaps run length	0	2	uncalibrated	train

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Method	Score	Threshold rule	Event policy	Merge gap	Params	Calibration	Tuned on
Linear autoencoder	linear projection residual	train quantile	merge gap	60	8		uncalibrated train
Regime-conditioned threshold	robust max abs z	regime conditioned quantile	merge gap	60	3		uncalibrated train
Robust change point	absolute first difference	train quantile	merge gap	60	1		uncalibrated train
SPOT	spot streaming excess	spot zero	merge gap	60	3		uncalibrated train
Negative-control region	negative control distance	train quantile	merge gap	60	4		uncalibrated train
Recurrence score	robust max abs z	train quantile	extremal index declustering	0	3		uncalibrated train
Failure-prototype region	negative prototype distance	train quantile	merge gap	60	4		uncalibrated train
Rare-state region	negative rare state distance	train quantile	merge gap	60	4		uncalibrated train

The complete benchmark including runtime, peak memory and configuration hashes is distributed as `event_benchmark_full.csv` in the reproducibility package; every value in the compact main-paper table is recomputable from it.

Table S4: Fairness audit for the event-level benchmark. Shared protocol elements are checked for identity; event conversion is reported rather than required to be uniform, because several methods exist to vary it.

check	status	detail
configuration_hash	pass	one configuration_hash per dataset across all methods
target_events	pass	one target_events per dataset across all methods
leakage_control	pass	one leakage_control per dataset across all methods
no_test_partition_tuning	pass	no method declares test-partition tuning
method_result_distinctness	reported	methods with identical results (reported, not hidden): metropt: empirical_horizon_risk, failure_prototype_region; metropt: conformal_anomaly, fixed_run_declustering, global_empirical_threshold; metropt2: conformal_anomaly, ferro_seggers_event_policy, fixed_run_declustering, global_empirical_threshold
event_conversion_declared	reported	event policies in use: extremal_index_declustering, ferro_seggers_run_length, fixed_run_declustering, k_gaps_run_length, merge_gap; merge gaps in use: 0, 60
declared_specifications	pass	every reported method has a declared specification

4 Additional Real-Data Results

MetroPT, MetroPT2, SCANIA Component X, Hydraulic Systems, and SECOM differ in sampling, entity structure, labels, and exposure. The source registry records acquisition status and provenance for each dataset. MetroPT and MetroPT2 are compressor telemetry resources, SCANIA Component X is a fleet-level truck dataset with anonymized component histories, Hydraulic Systems is an experimental test-rig cycle dataset, and SECOM is a semiconductor process/yield dataset [5–9]. Because the datasets support different estimands and independent units, the industrial evidence is presented as a limitation and reproducibility target rather than a completed population-level comparison.

The event timeline artifact is retained because it can reveal aligned-axis mistakes, alarm bursts outside failure windows, and threshold behavior that aggregate tables may hide. It does not replace event matching, calibration, or independent-unit uncertainty.

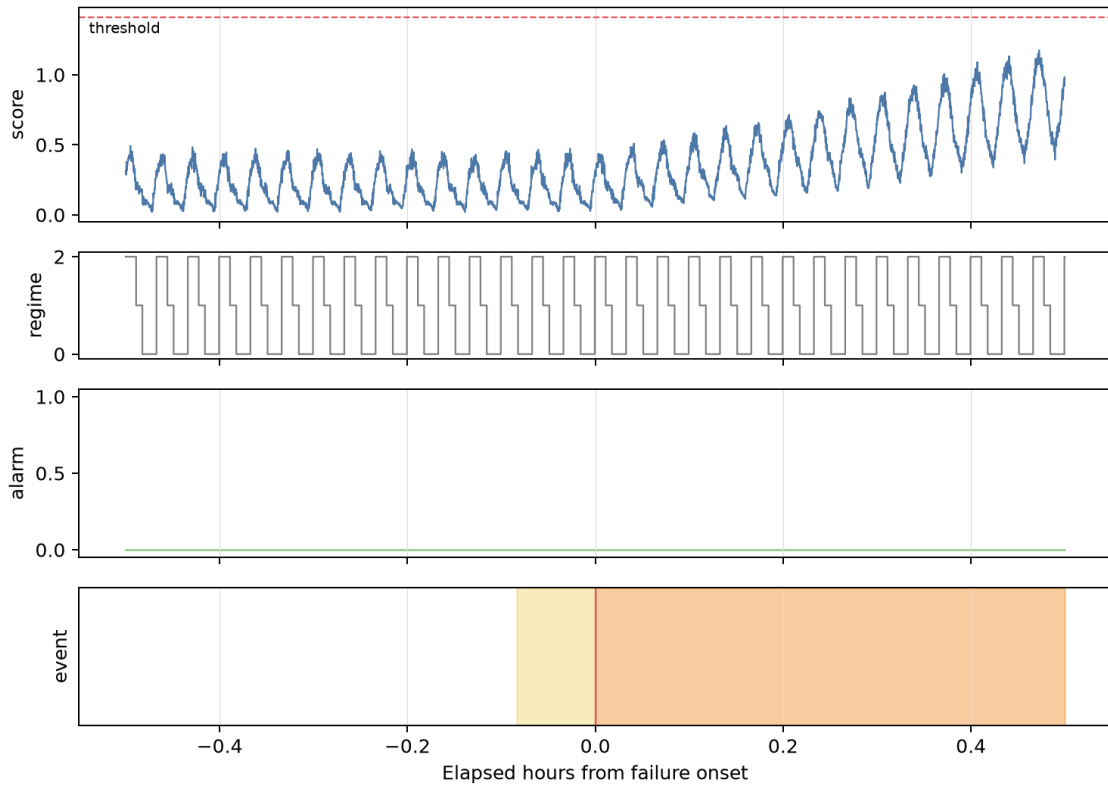


Figure S3: Supplementary alarm timeline generated from the cached processed data.

4.1 Global alarm burden for every method

The main paper shows the two-scale timeline for the registered recurrence score on each compressor dataset. Figures S4 and S5 extend the global scale to every method in the benchmark, binned identically and drawn on a shared vertical axis.

The shared axis is the point of these figures. Two methods can report similar episode counts and distribute those episodes very differently over the test split, and only the second property determines what an operator experiences. Reading down each column shows which methods alarm continuously, which alarm in bursts, and which produce nothing at all; the episode total printed on each strip ties the panel to the corresponding row of Table S2.

Only the registered method carries a per-sample local trace, because the local view exists to explain one detection and the per-sample series for nineteen methods on two datasets

is not a reviewable artifact. The binned global series for every method is distributed as `event_timeline_overview.csv`, and the local trace as `event_timeline_trace.csv`.

4.2 Non-event stress-test results

Table S5 gives the high-quantile diagnostic on Hydraulic Systems, SCANIA Component X, and SECOM. All three attain $F_1 = 0$. These are not hidden failures of the method: their independent units are cycles, vehicles, and wafers rather than compressor failure episodes, so the estimand differs from the one the main results address. They are evidence that a simple extreme-score adapter does not transfer across laboratory cycle states, vehicle-level repair risk, and wafer-level yield labels without redesigning the target and the evaluation, and they are reported here rather than in the main text for that reason.

Table S5: Non-event real-data diagnostic rows kept separate from event alarms.

Data	Unit	Units	Target	Pred	Recall	Precision	F1	Brier	Limit
Hydraulic load		441	431	0	0	0	0	0.977	cycle
cycle									labels
SCANIA vehicle		5,045	142	132	0	0	0	0.0543	estimand
									mis-
									match
SECOM wafer		314	17	10	0	0	0	0.086	yield
									target

4.3 Full decomposition and control tables

Table S6 is the complete score, threshold, cluster and alarm decomposition for all 38 dataset-method rows. The main text reports the failure layer derived from it; this is the underlying stage-by-stage evidence.

Table S6: Score, threshold, cluster and alarm decomposition per method. Exceedances are counts of samples, clusters are counts of extreme clusters, and the remaining columns are counts of alarm episodes; the columns are therefore not comparable across stages. Test observations are constant within a dataset (MetroPT 2,154,718; MetroPT2 1,423,388) and are omitted. Labelled failures are ground truth and are not part of this detector flow. The failure layer names the earliest stage at which the method’s evidence broke down.

Data	Method	Exceed.	Clusters	Episodes	Matched	False	Failure layer
MetroPT	Engineering threshold	320,251	2,583	2,583	1	2,579	no operationally acceptable result score
MetroPT	Best sensor threshold	24,870	2,268	2,268	1	2,266	
MetroPT	Global empirical	53,376	2,537	2,537	1	2,533	no operationally acceptable result score
MetroPT	Regime empirical	18,412	2,529	2,529	1	2,525	no operationally acceptable result score

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Data	Method	Exceed.	Clusters	Episodes	Matched	False	Failure layer
MetroPT	Classical POT/GPD	25,473	2,533	2,533	1	2,529	no operationally acceptable result
MetroPT	Fixed-run declustering	53,376	3,359	2,537	1	2,533	no operationally acceptable result
MetroPT	Ferro-Segers policy	53,376	3,359	2,554	1	2,550	no operationally acceptable result
MetroPT	K-gaps policy	53,376	3,359	2,553	1	2,549	no operationally acceptable result
MetroPT	SPOT-style	579	32	32	0	32	event-matching score
MetroPT	Isolation Forest	114,791	2,577	2,577	1	2,573	score
MetroPT	Robust changepoint	60,164	2,590	2,590	1	2,586	score
MetroPT	Linear autoencoder	108,683	4,721	4,721	1	4,715	no operationally acceptable result
MetroPT	Compact autoencoder	603,354	5,341	5,341	1	5,329	no operationally acceptable result
MetroPT	Conformal anomaly	53,376	2,537	2,537	1	2,533	no operationally acceptable result
MetroPT	Empirical horizon risk	0	0	0	0	0	score
MetroPT	Dynamical EVT score	53,376	3,359	2,770	1	2,766	no operationally acceptable result
MetroPT	Failure prototype	0	0	0	0	0	score
MetroPT	Rare-state region	14,511	892	892	1	888	score
MetroPT	Negative control	3,787	193	193	0	193	score
MetroPT2	Engineering threshold	302,635	614	614	1	613	no operationally acceptable result
MetroPT2	Best sensor threshold	101,276	354	354	1	350	no operationally acceptable result

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Data	Method	Exceed.	Clusters	Episodes	Matched	False	Failure layer
MetroPT2	Global empirical	37,583	882	882	1	878	no operationally acceptable result
MetroPT2	Regime empirical	36,123	2,361	2,361	1	2,357	no operationally acceptable result
MetroPT2	Classical POT/GPD	25,472	889	889	1	885	no operationally acceptable result
MetroPT2	Fixed-run declustering	37,583	1,552	882	1	878	no operationally acceptable result
MetroPT2	Ferro-Segers policy	37,583	1,552	882	1	878	no operationally acceptable result
MetroPT2	K-gaps policy	37,583	1,552	887	1	883	no operationally acceptable result
MetroPT2	SPOT-style	439	18	18	0	18	event-matching
MetroPT2	Isolation Forest	65,701	1,742	1,742	1	1,734	no operationally acceptable result
MetroPT2	Robust changepoint	29,349	1,357	1,357	1	1,350	score
MetroPT2	Linear autoencoder	250,352	1,157	1,157	1	1,156	no operationally acceptable result
MetroPT2	Compact autoencoder	32,795	783	783	1	779	no operationally acceptable result
MetroPT2	Conformal anomaly	37,583	882	882	1	878	no operationally acceptable result
MetroPT2	Empirical horizon risk	0	0	0	0	0	threshold
MetroPT2	Dynamical EVT score	37,583	1,552	893	1	889	no operationally acceptable result
MetroPT2	Failure prototype	233,643	3	3	1	2	insufficient independent evidence

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Data	Method	Exceed.	Clusters	Episodes	Matched	False	Failure layer
MetroPT2	Rare-state region	250,037	690	690	1	689	no operationally acceptable result score
MetroPT2	Negative control	30,329	689	689	0	689	

Table S7 reports the unconditioned matched-control comparison, with component metrics for each control family. The main text uses the detection-conditioned version, because comparing alarm burden without conditioning on detection rewards a region for not detecting; this table is retained so the component metrics behind that comparison remain inspectable.

4.4 Dataset roles and leakage audit

Table S8 records the empirical role, sampling structure, independent unit, independent positives, rows, and target type for every dataset. The independent-unit column is the quantity that governs which datasets can carry the primary claim.

Table S9 gives the terminal result of each registered leakage check: held-out failures excluded from feature construction, from scaling, and from regime inference, with thresholds and references fitted on the training split alone.

4.5 Matched controls under all conditions

Table eftab:detection-aware-controls gives every control family under all four conditions: all draws, detecting draws only, recall-matched draws and occupancy-matched draws. The main text reports the detecting-only rows, because conditioning on detection is what makes a burden comparison meaningful, but the unconditioned rows are what show how much the unconditioned comparison flattered the observed region, and the two matched conditions are the ones that preserve its structure while breaking its relation to the failure.

Table S10: Detection-aware matched controls, all conditions. Every comparison reports the share of control draws that detected the failure, because a burden advantage means nothing if the observed region did not detect while the controls did. Utility is the predeclared joint utility, which charges false alarms per operating day and warning exposure separately and pays a capped early-warning reward only when the failure was detected.

Data	Control family	Condition	Draws	Ctrl det. rate	Obs. de- tected	Obs. utility	Ctrl median	Percentile
MetroPT	episode label per- mutation	All draws	500	0	no	0	0	1
MetroPT	episode label per- mutation	Detecting only	0	0	no	0	Not es- timable	Not es- timable
MetroPT	episode label per- mutation	Recall matched	500	0	no	0	0	1
MetroPT	episode label per- mutation	Occupancy matched	500	0	no	0	0	1

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Data	Control family	Condition	Draws	Ctrl det. rate	Obs. detected	Obs. utility	Ctrl median	Percentile
MetroPT	phase randomised score	All draws	500	0	no	0	0	1
MetroPT	phase randomised score	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	phase randomised score	Recall matched	500	0	no	0	0	1
MetroPT	phase randomised score	Occupancy matched	500	0	no	0	0	1
MetroPT	prototype source permutation	All draws	500	0	no	0	0	1
MetroPT	prototype source permutation	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	prototype source permutation	Recall matched	500	0	no	0	0	1
MetroPT	prototype source permutation	Occupancy matched	500	0	no	0	0	1
MetroPT	random occupancy	All draws	500	0	no	0	0	1
MetroPT	random occupancy	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	random occupancy	Recall matched	500	0	no	0	0	1
MetroPT	random occupancy	Occupancy matched	500	0	no	0	0	1
MetroPT	regime matched rare region	All draws	500	0	no	0	0	1
MetroPT	regime matched rare region	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	regime matched rare region	Recall matched	500	0	no	0	0	1
MetroPT	regime matched rare region	Occupancy matched	500	0	no	0	0	1
MetroPT	time shifted prototype window	All draws	500	0	no	0	0	1
MetroPT	time shifted prototype window	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	time shifted prototype window	Recall matched	500	0	no	0	0	1

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Data	Control family	Condition	Draws	Ctrl det. rate	Obs. de- tected	Obs. utility	Ctrl median	Percentile
MetroPT	time shifted prototype window	Occupancy matched	500	0	no	0	0	1
MetroPT2	episode label per- mutation	All draws	500	0.17	yes	103	-1.82	1
MetroPT2	episode label per- mutation	Detecting only	85	0.17	yes	103	103	1
MetroPT2	episode label per- mutation	Recall matched	85	0.17	yes	103	103	1
MetroPT2	episode label per- mutation	Occupancy matched	500	0.17	yes	103	-1.82	1
MetroPT2	phase randomised score	All draws	500	1	yes	103	58.5	1
MetroPT2	phase randomised score	Detecting only	500	1	yes	103	58.5	1
MetroPT2	phase randomised score	Recall matched	500	1	yes	103	58.5	1
MetroPT2	phase randomised score	Occupancy matched	500	1	yes	103	58.5	1
MetroPT2	prototype source per- mutation	All draws	500	0.068	yes	103	-1.82	0.986
MetroPT2	prototype source per- mutation	Detecting only	34	0.068	yes	103	103	0.794
MetroPT2	prototype source per- mutation	Recall matched	34	0.068	yes	103	103	0.794
MetroPT2	prototype source per- mutation	Occupancy matched	429	0.068	yes	103	-1.82	0.984
MetroPT2	random occupancy	All draws	500	0.054	yes	103	-1.82	0.986
MetroPT2	random occupancy	Detecting only	27	0.054	yes	103	103	0.741
MetroPT2	random occupancy	Recall matched	27	0.054	yes	103	103	0.741
MetroPT2	random occupancy	Occupancy matched	449	0.054	yes	103	-1.82	0.984
MetroPT2	regime matched rare region	All draws	500	0.872	yes	103	65.1	1
MetroPT2	regime matched rare region	Detecting only	436	0.872	yes	103	65.3	1

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Data	Control family	Condition	Draws	Ctrl det. rate	Obs. de- tected	Obs. utility	Ctrl median	Percentile
MetroPT2	regime matched rare region	Recall matched	436	0.872	yes	103	65.3	1
MetroPT2	regime matched rare region	Occupancy matched	452	0.872	yes	103	65.2	1
MetroPT2	time shifted prototype window	All draws	500	0.174	yes	103	-1.82	1
MetroPT2	time shifted prototype window	Detecting only	87	0.174	yes	103	103	1
MetroPT2	time shifted prototype window	Recall matched	87	0.174	yes	103	103	1
MetroPT2	time shifted prototype window	Occupancy matched	500	0.174	yes	103	-1.82	1

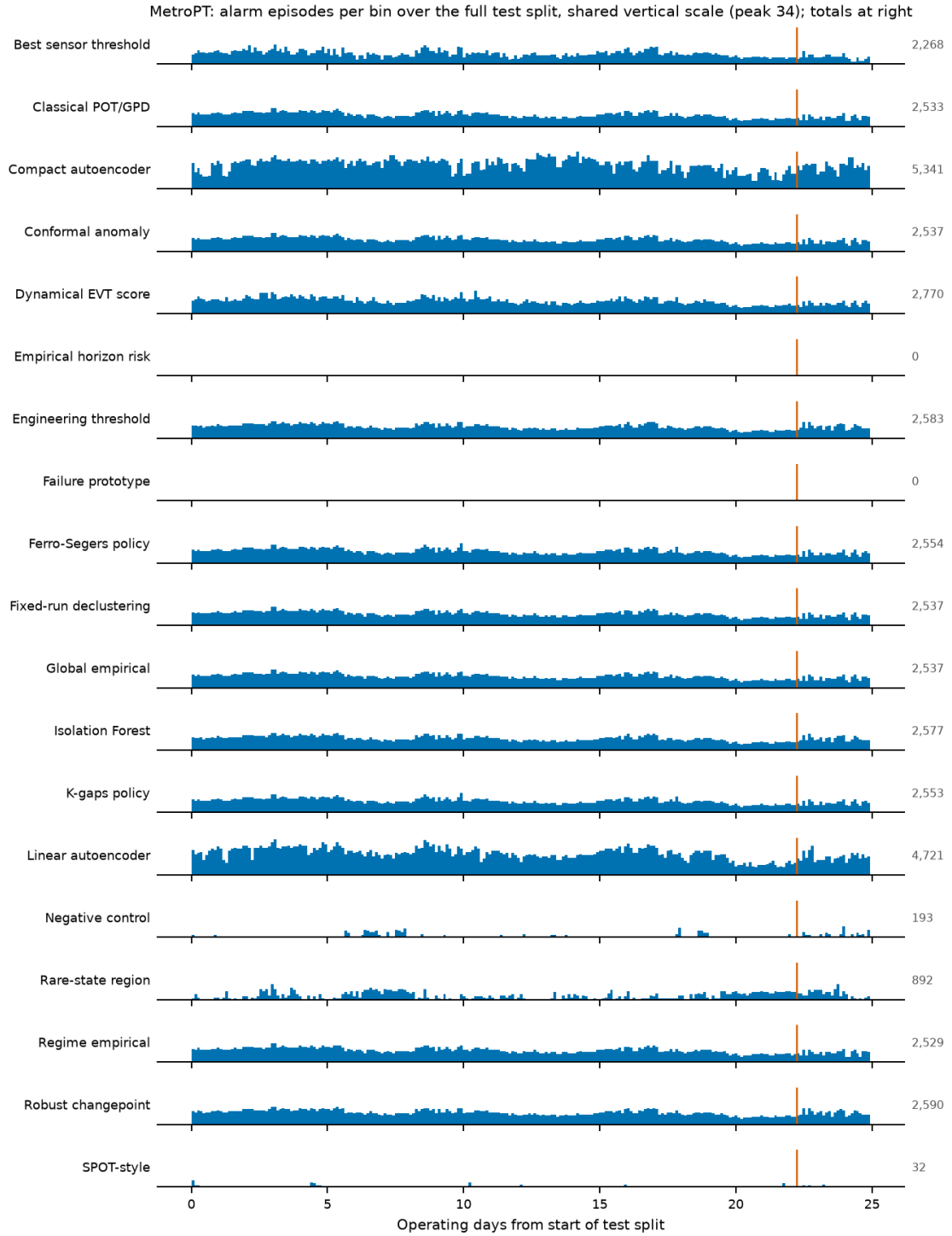


Figure S4: MetroPT global alarm burden for every benchmark method over the held-out test split, on a shared vertical scale. Vertical rules mark the labelled failure onset; the count on each strip is that method's total alarm episodes.

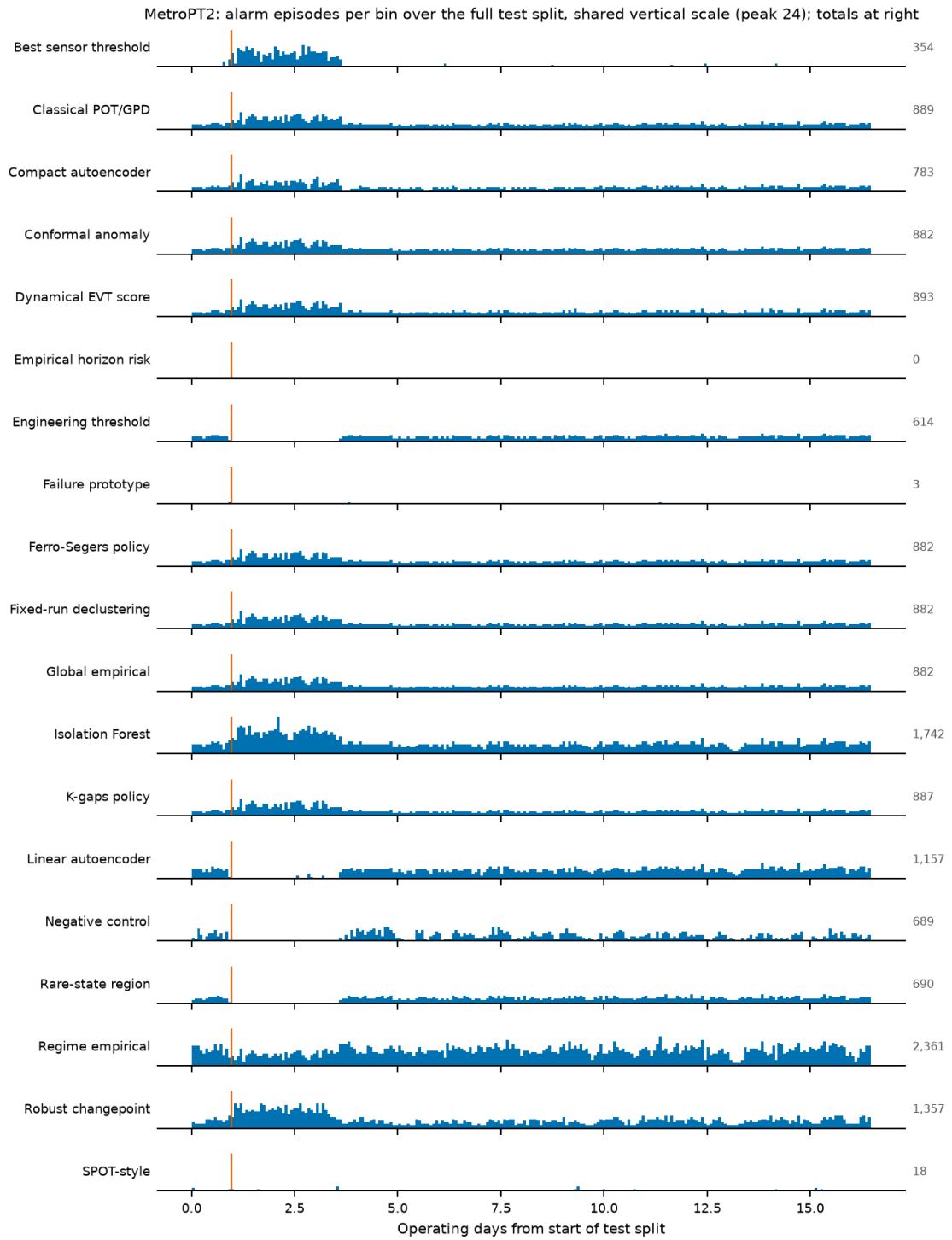


Figure S5: MetroPT2 global alarm burden for every benchmark method over the held-out test split, on a shared vertical scale. Vertical rules mark the labelled failure onset; the count on each strip is that method's total alarm episodes.

Table S7: Observed failure-prototype alarm burden against 500 matched-control draws per family.

Data	Observed	Control	Observed FA/day	Control median	95% interval	Percentile	Interpretation
MetroPT	Failure prototype	Label permuta- tion	0	0	0-0	1	not lower burden
MetroPT	Failure prototype	Phase- randomised	0	0	0-0	1	not lower burden
MetroPT	Failure prototype	Prototype permuta- tion	0	0	0-0	1	not lower burden
MetroPT	Failure prototype	Random occu- pancy	0	0	0-0	1	not lower burden
MetroPT	Failure prototype	Rare- region matched	0	0	0-0	1	not lower burden
MetroPT	Failure prototype	Time- shifted	0	0	0-0	1	not lower burden
MetroPT2	Failure prototype	Label permuta- tion	0.121	0.182	0.121- 0.243	0.062	lower alarm burden than controls
MetroPT2	Failure prototype	Phase- randomised	0.121	41.8	41.6-41.9	0	lower alarm burden than controls
MetroPT2	Failure prototype	Prototype permuta- tion	0.121	0.182	0.0607- 0.182	0.296	lower alarm burden than controls
MetroPT2	Failure prototype	Random occu- pancy	0.121	0.182	0.0607- 0.182	0.336	lower alarm burden than controls
MetroPT2	Failure prototype	Rare- region matched	0.121	34.6	33.4-40.3	0	lower alarm burden than controls
MetroPT2	Failure prototype	Time- shifted	0.121	0.182	0.121- 0.243	0.058	lower alarm burden than controls

Table S8: Dataset roles and evidential units.

Dataset	Role	Sampling	Unit	Ind. positives	Rows	Target
Hydraulic	cycle condition	test-rig cycles	load cycle	431	2,205	condition states
MetroPT	event warning	ordered telemetry	failure episode / operating day, not timestamp	1	10,773,588	failure labels
MetroPT2	event warning	ordered telemetry	failure episode / operating day, not timestamp	1	7,116,940	failure labels
SCANIA	repair-risk ranking	vehicle histories	vehicle	142	1,516,819	published labels and time-to- event files
SECOM	yield failure	wafer rows	wafer / production example	17	1,567	yield labels

Table S9: Terminal leakage-audit results for the registered workflow.

Check	Result	Evidence
Held-out failures excluded from feature construction	passed	causal trailing windows
Held-out failures excluded from scaling	passed	train-split robust scaling
Held-out failures excluded from regime inference	passed	unsupervised labels excluded
Held-out failures excluded from target-region construction	passed	train-only prototypes
Test labels excluded from threshold selection	passed	train/calibration quantiles
Test labels excluded from lag selection	passed	registered lag grids
Test labels excluded from merge-gap selection	passed	frozen evaluation protocol
Test labels excluded from calibration	passed	validation-only score mapping
Maintenance boundaries respected	passed	boundary fields in protocol
SCANIA vehicles disjoint across partitions	passed	preparation raises on overlap
No future information in causal windows	passed	current and trailing observations only

5 Robustness and Negative Results

The robustness policy is to preserve unfavorable outcomes. Threshold sensitivity, run-length sensitivity, regime-definition changes, dangerous-region definitions, missingness perturbations, and noise perturbations are all represented as registered families in the generated ablation table. Some cells are summaries of planned or non-estimable effects rather than completed industrial contrasts, so the ablation table is treated as a coverage record rather than a positive performance result.

Negative and null findings are part of the evidence. The manuscript does not turn diagnostic evidence into broad industrial performance claims, and it does not use empirical reliability diagnostics as calibrated probability evidence. These constraints are enforced by the paper-check command and by generated audit artifacts.

5.1 Full Threshold and Alarm-Conversion Grids

Table S11: Failed-detection grid for threshold, recall, precision, and false-alarm burden.

threshold quantile	threshold	event recall	event precision	false alarm events per day	median warning lead time
0.9	1.119	0	0	168.5	
0.95	1.329	0	0	146.9	
0.97	1.388	0	0	133.9	
0.98	1.411	0	0	146.9	
0.99	1.439	0	0	138.2	

Table S12: Threshold and run-length stability diagnostics.

threshold quantile	run length	n exceedances	n clusters
0.9	0	2000	90
0.9	2	2000	41
0.9	5	2000	39
0.9	10	2000	38
0.9	20	2000	38
0.95	0	1000	105
0.95	2	1000	43
0.95	5	1000	34
0.95	10	1000	33
0.95	20	1000	33
0.97	0	600	141
0.97	2	600	44
0.97	5	600	31
0.97	10	600	31
0.97	20	600	31
0.98	0	400	157
0.98	2	400	48
0.98	5	400	34
0.98	10	400	31
0.98	20	400	31
0.99	0	200	128
0.99	2	200	68
0.99	5	200	32

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threshold quantile	run length	n exceedances	n clusters
0.99	10	200	31

Table S13: Full alarm-conversion grid for merge gap and post-onset matching tolerance.

merge	tol after	recall	prec	F1	false alarms	dup	FA/day	lead	warn time
0	0	0	0	0	157	0	678.2		400
0	30	0	0	0	157	0	678.2		400
0	60	0	0	0	157	0	678.2		400
0	120	0	0	0	157	0	678.2		400
0	300	0	0	0	157	0	678.2		400
5	0	0	0	0	34	0	146.9		586
5	30	0	0	0	34	0	146.9		586
5	60	0	0	0	34	0	146.9		586
5	120	0	0	0	34	0	146.9		586
5	300	0	0	0	34	0	146.9		586
20	0	0	0	0	31	0	133.9		604
20	30	0	0	0	31	0	133.9		604
20	60	0	0	0	31	0	133.9		604
20	120	0	0	0	31	0	133.9		604
20	300	0	0	0	31	0	133.9		604
60	0	0	0	0	31	0	133.9		604
60	30	0	0	0	31	0	133.9		604
60	60	0	0	0	31	0	133.9		604
60	120	0	0	0	31	0	133.9		604
60	300	0	0	0	31	0	133.9		604

5.2 Full Score-Stratification Bins

Table S14: Split-aware score-stratification intervals using training-ranked scores.

split	lower	upper	count	pos	mean rank	obs prop	CI low	CI high
test	0.9	1	4000	401	1	0.1003	0.09132	0.1099
train	0	0.1	1199	0	0.05	0	0	0.003194
train	0.1	0.2	1200	0	0.15	0	0	0.003191
train	0.2	0.3	1201	0	0.25	0	0	0.003188
train	0.3	0.4	1199	0	0.35	0	0	0.003194
train	0.4	0.5	1200	0	0.45	0	0	0.003191
train	0.5	0.6	1201	0	0.55	0	0	0.003188
train	0.6	0.7	1200	0	0.65	0	0	0.003191
train	0.7	0.8	1199	0	0.75	0	0	0.003194
train	0.8	0.9	1200	0	0.85	0	0	0.003191
train	0.9	1	1201	0	0.95	0	0	0.003188
val	0	0.1	215	52	0.04765	0.2419	0.1895	0.3033
val	0.1	0.2	234	47	0.1494	0.2009	0.1545	0.2568
val	0.2	0.3	219	64	0.2529	0.2922	0.236	0.3556
val	0.3	0.4	280	126	0.352	0.45	0.3928	0.5086
val	0.4	0.5	292	86	0.4522	0.2945	0.2452	0.3492
val	0.5	0.6	283	119	0.5512	0.4205	0.3644	0.4787
val	0.6	0.7	366	197	0.651	0.5383	0.487	0.5887

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split	lower	upper	count	pos	mean rank	obs prop	CI low	CI high
val	0.7	0.8	323	164	0.7488	0.5077	0.4534	0.5618
val	0.8	0.9	317	134	0.8504	0.4227	0.3696	0.4777
val	0.9	1	1471	1311	0.9883	0.8912	0.8743	0.9061

5.3 Ablation Configuration Registry

Table S15: Ablation matrix summary.

ablation family	setting	summary value
threshold	0.9	0.1
threshold	0.95	0.05
threshold	0.97	0.03
threshold	0.98	0.02
threshold	0.99	0.01
run_length	0	157
run_length	2	48
run_length	5	34
run_length	10	31
run_length	20	31
lag	0	
lag	1	
lag	2	
lag	5	
lag	10	
regime	off	
regime	loaded	
regime	recovery	
dangerous_region	prototype	
dangerous_region	constraint	
dangerous_region	density	
noise_missingness	none	
noise_missingness	noise_0.05	
noise_missingness	missing_0.05	

5.4 Baseline Runner Smoke Audit

Table S16: Synthetic smoke results for the shared baseline runner.

Baseline	Validation F1	Runtime (s)	Parameters
Engineering threshold	0	0.00142	12
Global empirical	0.911	0.00279	1
Regime empirical	0.892	0.0111	3
POT/GPD	0.911	1.77	3
Fixed-run declustering	0.904	0.00892	2
K-gaps declustering	0.904	0.00987	2
SPOT-style	0.919	14.8	4
Isolation Forest	0.92	2.63	120
Robust changepoint	0.566	0.0648	12
Linear autoencoder	0.955	0.0259	54
Conformal anomaly	0.319	0.0162	200
Empirical horizon risk	0.919	0.00306	200

6 Reproducibility

The build is organized around reproducible artifacts. The root command `make paper-assets` regenerates figures, tables, result macros, the results synthesis document, and per-file provenance records. The command `make paper` builds the main manuscript after regenerating assets. The command `make supplement` builds this supplement after the same asset step. The command `make paper-check` compiles both documents and audits generated assets, citations, compiled PDFs, and source markers. The command `make submission-package` assembles the reviewed manuscript package, reviewer report, revision matrix, availability statements, protocol, lock file, and artifact manifests without copying raw third-party datasets.

The generated macros provide stable result values used by the manuscript and supplement. Exact build identifiers and file-integrity records are stored in the repository manifests rather than printed in the narrative text.

Machine-readable interpretation and citation audits are written to `paper/generated/` by the paper checker. Manual edits to generated figures or tables are detected through the artifact-manifest audit.

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